WEEK 4 · LESSON 24 OF 24 · HANDOUT

Final Project: AI Problem Analysis and Certificate

Study time: **about 90 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=hQX_wIW9Nh0 — open the lesson online to watch inside the course.

Your goal in this lesson: Produce a complete, professional AI problem analysis that brings together all four weeks — and earn your AI Foundations certificate.

What you will do

Choose one real problem in Rwanda or your community that AI could help with. Write an **AI Problem Analysis** of 4–6 pages (or 12–15 slides). Build on your Week 1, 2 and 3 projects — you have already done much of the thinking.

This is the kind of document a consultant, analyst or founder writes before any money is spent on building. It is also a strong first item for your portfolio.

Part 1 — The problem (Week 1)

- Who has the problem, how big is it, and how is it handled today?
- Describe the AI system as an **agent**: write its **PEAS** table and classify its **environment**.
- Name the **problem type** (classification, regression, clustering, generation…).

Part 2 — The learning plan (Week 2)

- **Learning type** — supervised, unsupervised or reinforcement, and why.
- **Data** — features and label, sources, quality risks, leakage check, and whether the data is **representative**.
- **Workflow** — a short plan for each of the seven stages, including deployment and monitoring for **drift**.
- Could a simple rule do the job instead? Justify using AI.

Part 3 — Model and evaluation (Week 3)

- A **baseline** method and a stronger method, with the trade-off.
- How you will split data (training, validation, test).
- The **main measure** (accuracy, precision, recall, MAE…) and why — which mistake is more costly?
- A numeric **success target**.

Part 4 — Responsible AI (Week 4)

- **Fairness** — which groups could get worse results, and which per-group checks you will run.
- **Explainability** — how will a person understand or challenge a decision?
- **Privacy and security** — which personal data is used, how you will follow Rwanda's data-protection law, and one security threat with a protection.
- **Governance** — the risk level (using the EU AI Act categories as a guide), who is accountable, and how users can reach a human.

Part 5 — Recommendation

One page: should this project go ahead? What is the first small step (a pilot)? What would make you stop?

Marking guide (self-check)

Area	What "excellent" looks like
Problem framing	Clear user, decision and numeric success measure
Technical correctness	Correct agent, problem type, learning type and measures
Data thinking	Specific features, honest quality and representativeness risks
Responsibility	Concrete fairness, privacy and governance steps — not general words
Communication	Simple English, clear structure, a busy reader understands quickly

Getting your certificate

1. Complete **every lesson** in the course (tick each one as done).
2. Finish your final project and keep it in your portfolio.
3. When all lessons are complete, your **WAMUGII Certificate of Completion** unlocks on the course page and you receive a congratulations email. Download it and add it to your CV and LinkedIn.

What you have achieved

You can now explain AI precisely, describe any AI system as an agent, frame problems correctly, understand how models learn and are evaluated, and judge AI's impact on people and society. That foundation sets you apart from people who only *use* AI tools without understanding them.

Remember this: A strong AI problem analysis connects the problem, the data, the method, honest evaluation and responsible use — in clear language anyone can follow.

Name: _____ Date: _____

Activity

Task 1

Complete Parts 1–5. Ask one person from the field you chose (a teacher, nurse, farmer, trader or official) to read your recommendation and tell you what they think. Add their feedback as a final short section — real feedback is what turns a school project into professional work.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does Part 1 of the final project cover?
   - A. Only the code
   - B. The problem, the agent's PEAS table, its environment and the problem type
   - C. Only the budget
   - D. The certificate design
2. Why include a baseline method in your analysis?
   - A. To have a reference point a stronger method must beat
   - B. It is not needed
   - C. Because baselines are always best
   - D. To fill pages
3. Which part of the project checks fairness, privacy and governance?
   - A. None
   - B. Part 3
   - C. Part 4 — Responsible AI
   - D. Part 1
4. What should the recommendation include?
   - A. Only praise for AI
   - B. A list of programming languages
   - C. The model's source code
   - D. Whether to go ahead, a first small pilot step, and what would make you stop
5. How is the AI Foundations certificate issued?
   - A. It unlocks on the course page when every lesson is completed
   - B. Only after paying a fee
   - C. By posting a letter
   - D. It is not available
6. What does 'excellent' communication look like in the marking guide?
   - A. Only tables
   - B. Long, complex sentences with many technical words
   - C. Simple English and clear structure that a busy reader understands quickly
   - D. No headings at all

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